

Scenario 1 – Department Salary Analysis

Question: The HR department wants to identify departments where the average employee salary is greater than the company's salary benchmark. Display the Department and its Average Salary. Sort the result by Average Salary in descending order and display only the top 3 departments.

Answer (MySQL):

```
SELECT Department, AVG(Salary) AS Average_Salary
FROM Employee
GROUP BY Department
HAVING AVG(Salary) > (SELECT AVG(Salary) FROM Employee)
ORDER BY Average_Salary DESC
LIMIT 3;
```

Scenario 2 – Top Performing Products

Question: Display each Product Category along with the total quantity sold. Only display categories that have sold more than the company's minimum sales requirement. Sort the result from highest to lowest quantity sold and display only the top 5 categories.

Answer (MySQL):

```
SELECT Category, SUM(Quantity) AS Total_Quantity
FROM Sales
GROUP BY Category
HAVING SUM(Quantity) > 100
ORDER BY Total_Quantity DESC
LIMIT 5;
```

Scenario 3 – Customer Purchase Report

Question: Display only customers whose purchase amount satisfies the company's purchase criteria. Remove duplicate city names in the report and display the customers in ascending alphabetical order.

Answer (MySQL):

```
SELECT Customer_Name, City, Purchase_Amount  
FROM Customer  
WHERE Purchase_Amount > 500  
ORDER BY Customer_Name ASC;
```

```
SELECT DISTINCT City  
FROM Customer;
```

Scenario 4 – Hospital Department Patient Analysis

Question: Display the Department Name and Total Patients. Show only those departments that have treated more than the hospital's required minimum number of patients. Sort the result based on the total number of patients in descending order.

Answer (MySQL):

```
SELECT Department, COUNT(*) AS Total_Patients
FROM Patient
GROUP BY Department
HAVING COUNT(*) > 5
ORDER BY Total_Patients DESC;
```

Scenario 5 – Employee Age and Salary Report

Question: Display the employee details who satisfy both age and salary criteria, arrange the report by Salary in descending order, and return only the top 5 employees.

Answer (MySQL):

```
SELECT *  
FROM Employee  
WHERE Age > 25 AND Salary > 50000  
ORDER BY Salary DESC  
LIMIT 5;
```